An Internship Report

Of

Data Science Master Virtual Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B-Tech (CSE)

By

Roll Number of Student:- 2410031286
Name of Student: HARSH PATEL

Batch: 2026-27

CANDIDATE'S DECLARATION

I, HARSH PATEL, do hereby declare that the internship report titled "**Data Science Master Virtual Internship**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: HARSH PATEL

Roll No.: 2410031286

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the **AICTE – EduSkills Virtual Internship Program team and EduSkills Academy** for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of **Data Science Master**. The program, supported by **Siemens**, offered an exceptional curriculum bridging academic learning with real-world industry practices in data engineering, machine learning, and enterprise platform administration. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:

Student Name: HARSH PATEL

Roll No.: 2410031286

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3. INTERNSHIP COMPLETION CERTIFICATE



Internship Offer Letter

EduSkills®
Nation Building Through Skills

Ref No: C16Y2026M081395968

Internship Offer Letter

Student Name	: Harsh Patel
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Siemens Data Science Master
Internship Duration	: April 2026 - June 2026
AICTE Student ID	: STU698c933615c991770820406

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 8-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program, in the domain of **"Data Science Master"**, supported by **Siemens**. The internship was carried out from April 2026 to June 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to bridge the gap between theoretical understanding of data science and machine learning and their real-world, production-grade deployment. Over the course of the programme, the learner was exposed to industry-relevant tools and practices used to take data science projects from problem identification and data engineering all the way to model deployment, enterprise platform administration, and monitoring.

The internship followed a structured, week-wise curriculum combining conceptual learning with hands-on assignments using RapidMiner, weekly assessments, project documentation, and a final credential validation. On successful completion of all the requirements, including the final grade point assessment, a Virtual Internship Certificate was awarded by EduSkills Academy with a Grade of **O (Outstanding)**.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified organization working under the theme "Nation Building Through Skills". EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal, which operates under the Ministry of Education, Government of India. The organization collaborates with leading technology companies and academic institutions to design industry-aligned virtual internship curricula that help engineering students strengthen their employability and gain practical exposure to emerging technology domains.

The Data Science Master programme is supported by **Siemens**, a global leader in industrial automation, digitalization, and enterprise software. Siemens brings its

industry expertise to the programme, ensuring that the curriculum reflects real-world tools, workflows, and platform practices that students are expected to encounter in professional data science roles. The programme is administered through the **RapidMiner AI Hub**, an enterprise-grade machine learning and data science platform.

4.3 Problem Statement

Although a large number of engineering students learn foundational statistics and programming as part of their academic coursework, very few receive structured, hands-on exposure to the complete data science lifecycle as practiced in industry. Building a machine learning model in a Jupyter Notebook is only one component of real-world data science; designing automated data preparation pipelines, selecting and tuning production-grade models, deploying and monitoring them on enterprise platforms, and administering AI infrastructure represent equally important — and typically neglected — skills.

The problem addressed by this internship was to understand and practically implement the full data science project lifecycle: from identifying AI-suitable business problems and engineering raw data, through applying foundational and advanced machine learning techniques, to deploying and monitoring models and administering an enterprise-scale AI hub. The internship used RapidMiner as the primary platform to give learners industry-standard, visual, and code-complementary exposure to these workflows.

4.4 Project Objectives

- To understand how to identify business problems that are well-suited for AI and machine learning solutions, and to scope data science projects professionally.
- To learn how to access, load, transform, and calculate data using industry tools, establishing sound data engineering practices as the foundation for ML workflows.
- To build foundational machine learning skills, including classification, regression, model evaluation, and performance metric interpretation.

- To master advanced data handling techniques including missing value treatment, outlier management, feature engineering, and automated data preparation pipelines.
- To learn and apply advanced machine learning techniques, including ensemble methods, hyperparameter tuning, model selection, and cross-validation strategies.
- To gain hands-on experience in deploying and monitoring machine learning models in production, including live inference, performance tracking, and retraining triggers.
- To configure and manage the RapidMiner AI Hub, including user administration, resource governance, workflow scheduling, and enterprise platform operations.
- To apply all the above concepts in a real-world, end-to-end capstone project demonstrating the complete data science lifecycle from problem scoping to enterprise deployment.

4.5 Scope of the Project

The scope of this internship spans the entire data science project lifecycle — from identifying and scoping AI-suitable business problems, engineering and preparing raw data, applying foundational and advanced machine learning techniques, deploying and monitoring models in production, to administering an enterprise AI platform. The curriculum progressed from professional-level foundations in the first three weeks to master-level competencies in the subsequent five weeks, culminating in a Final Credential Validation and Internship Grade Point Assessment.

The internship was conducted in a virtual, simulation and lab-based environment provided by EduSkills Academy using the RapidMiner platform, supported by Siemens. It did not involve deployment on any live organizational production system. All practical exercises were completed within the structured weekly learning modules provided through the AICTE – EduSkills portal.

4.6 Technologies and Tools Used

- Programming & Data Analysis: **Python** — data manipulation, scripting, and ML model development
- ML & Data Science Platform: **RapidMiner Studio** — visual ML workflow design, model training, and evaluation
- Enterprise AI Platform: **RapidMiner AI Hub** — model deployment, monitoring, user administration, and governance
- Data Engineering: ETL pipelines, data transformation operators, feature engineering within RapidMiner
- Machine Learning Techniques: Classification, Regression, Ensemble Methods (Random Forest, Boosting), Hyperparameter Optimization, Cross-validation
- Model Deployment & Monitoring: Scheduled model execution, real-time scoring, drift detection, and retraining workflows
- Platform Administration: RapidMiner AI Hub configuration, user and role management, resource governance, and security settings
- Industry Support: **Siemens** — curriculum design, industry alignment, and enterprise use-case framing

4.7 System Architecture

The general architecture followed during the internship for developing and deploying a data science solution can be summarized in the following layers:

- **Business Problem Layer:** AI and ML use-case identification, problem scoping, and KPI definition to frame data science projects with clear business objectives.
- **Data Engineering Layer:** Data access, loading, transformation, and calculation using Python and RapidMiner ETL operators to prepare structured, analysis-ready datasets.
- **Machine Learning Layer:** Model training using RapidMiner Studio workflows — encompassing classification, regression, ensemble models, and hyperparameter tuning with rigorous cross-validation.

- **Deployment & Scoring Layer:** Production deployment of trained models on RapidMiner AI Hub for real-time and scheduled scoring, with performance monitoring dashboards.
- **Observability & Drift Management Layer:** Continuous tracking of deployed model accuracy and data drift, with retraining triggers and automated retraining pipelines to sustain model performance.
- **Platform Administration Layer:** Enterprise-level management of the RapidMiner AI Hub — user roles, resource allocation, workflow governance, and security compliance.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on RapidMiner labs, weekly assessments, and project submissions, as summarized in the table below.

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning. Understand use-case scoping, project framing, and the data science project lifecycle.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data using Python and RapidMiner ETL operators on real-world datasets to produce analysis-ready inputs.
Week 3	Machine Learning Professional	Build foundational machine learning skills including classification, regression, model selection, evaluation metrics, and performance interpretation using RapidMiner Studio.
Week 4	Data Preparation Master	Master advanced data handling and automation: missing value treatment, outlier detection, feature engineering, and building automated data preparation pipelines.
Week 5	Machine Learning Master	Learn advanced machine learning techniques: ensemble methods, hyperparameter optimization, model comparison, cross-validation strategies, and best-practice model selection.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models in production environments. Apply advanced use-case frameworks to industry datasets and evaluate models against real business KPIs.
Week 7	Platform Administration Master	Configure and manage the RapidMiner AI Hub: user management, resource governance, workflow scheduling, connection administration, and hub security settings.
Week 8	Platform Administration Master	Enterprise-level platform operations: advanced hub administration, scalability configuration, security compliance, and Final Credential Validation / Internship Grade Point Assessment.

Each week concluded with a weekly assessment to evaluate conceptual understanding, and selected modules required submission of project documentation and hands-on lab assignments. In the final week, a Final Credential Validation and Grade Point Assessment was conducted, and the overall grade was determined based on performance across weekly assessments, project submissions, and the final evaluation.

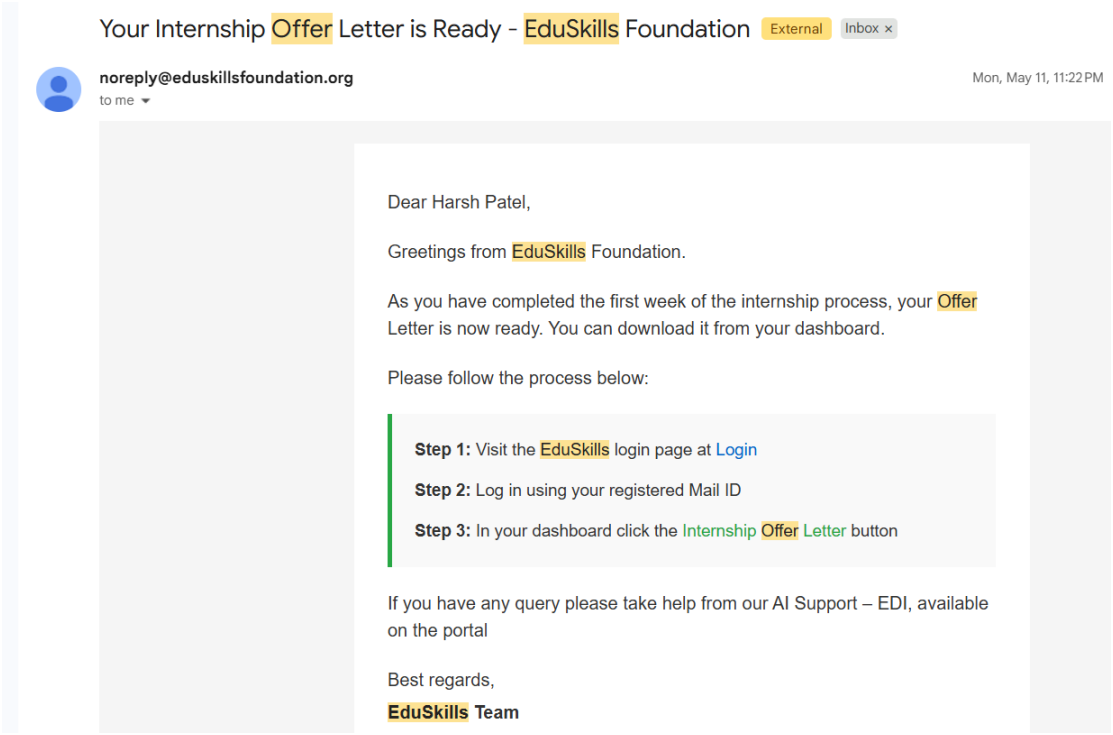
4.9 Expected Outcomes

- Ability to independently identify and scope business problems suited for AI and machine learning solutions in real enterprise contexts.
- Proficiency in data engineering practices: accessing, loading, transforming, and preparing raw datasets for downstream ML workflows using Python and RapidMiner.
- Foundational and advanced machine learning competency: training, evaluating, and optimizing classification and regression models with ensemble methods and hyperparameter tuning.
- Practical understanding of model deployment and monitoring: deploying trained models on enterprise platforms, tracking performance, and handling drift and retraining.
- Hands-on skill in configuring and administering the RapidMiner AI Hub for enterprise use, including user management, governance, resource allocation, and security compliance.
- A completed, end-to-end capstone experience demonstrating the full data science lifecycle from problem scoping through enterprise platform deployment and administration.
- Award of the Virtual Internship Certificate with **Grade O (Outstanding)** by EduSkills Academy, recognized under AICTE – National Internship Portal, Ministry of Education.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter (Ref No.: STU698c933615c991770820406) and the Internship Completion Certificate (Certificate ID: 41dd97631fc3c7baca42) issued by EduSkills Academy, confirming the successful completion of the 8-week AICTE – EduSkills Virtual Internship Program in Data Science Master, supported by Siemens,

are attached in Chapter 3 of this report. The certificate was awarded with **Grade O (Outstanding)**, corresponding to a score in the range of 90–100.



Your Internship Offer Letter is Ready - EduSkills Foundation External Inbox ×

noreply@eduskillsfoundation.org
to me

Mon, May 11, 11:22 PM

Dear Harsh Patel,

Greetings from EduSkills Foundation.

As you have completed the first week of the internship process, your Offer Letter is now ready. You can download it from your dashboard.

Please follow the process below:

Step 1: Visit the EduSkills login page at Login

Step 2: Log in using your registered Mail ID

Step 3: In your dashboard click the Internship Offer Letter button

If you have any query please take help from our AI Support – EDI, available on the portal

Best regards,
EduSkills Team

5. BIBLIOGRAPHY/REFERENCES

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